Nonnegative and positive factorizations

The Rowland–Wu polynomial identity for Sinkhorn limits

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Topic: Exact matrix scaling and balancing.

Rating rationale: Challenging because the full coefficient formula requires a general algebraic-combinatorial identity; community importance concerns exact formulas for a widely used matrix-scaling limit.

Problem statement

Let $m, n \geq 1$ and $A \in \mathbb{R}^{m\times n}_{>0}$. Let $\mathrm{Sink}(A)$ be its Sinkhorn limit with row sums 1 and column sums m/n: equivalently, the unique matrix $D_1 A D_2$ with these sums, where D_1, D_2 are positive diagonal matrices. Set $x = \mathrm{Sink}(A)_{11}$.

Define

$$\mathcal{D} = \{(R, C) : R \subseteq \{2, \dots, m\},\ C \subseteq \{2, \dots, n\},\ |R| = |C|\}.$$

Submatrix indices are in increasing order; empty determinants and products equal 1. Put

$$\Delta(R, C) = \det A_{\{1\}\cup R, \{1\}\cup C}, \qquad \Gamma(R, C) = a_{11} \det A_{R,C},$$

$$M(\mathcal{S}) = \prod_{(R,C)\in\mathcal{S}} \Delta(R, C) \prod_{(R,C)\in\mathcal{D}\setminus\mathcal{S}} \Gamma(R, C).$$

For a finite set U of integers and $s \in U$, let $p_s(U)$ be the position of s in the increasing list of U, starting at 1. For each $\mathcal{S} = \{(R_i, C_i) : 1 \leq i \leq k\} \subseteq \mathcal{D}$, define $H_{\mathcal{S}} \in \mathbb{Z}^{k\times k}$ by

$$(H_{\mathcal{S}})_{ii} = |R_i|(m + n) - mn, \qquad \tau_{ij} = (R_i \setminus R_j, R_j \setminus R_i, C_i \setminus C_j, C_j \setminus C_i),$$

and, for $i \neq j$,

$$(H_{\mathcal{S}})_{ij} = \begin{cases} (-1)^{p_s(R_j)+p_t(C_j)} m, & \tau_{ij} = (\emptyset, \{s\}, \emptyset, \{t\}), \\ (-1)^{p_s(R_i)+p_t(C_i)+1} n, & \tau_{ij} = (\{s\}, \emptyset, \{t\}, \emptyset), \\ (-1)^{p_s(C_i)+p_t(C_j)} m, & \tau_{ij} = (\emptyset, \emptyset, \{s\}, \{t\}), \\ (-1)^{p_s(R_i)+p_t(R_j)} n, & \tau_{ij} = (\{s\}, \{t\}, \emptyset, \emptyset), \\ 0, & \text{otherwise.} \end{cases}$$

The determinant is independent of the ordering chosen for $\mathcal{S}$. Is the following identity valid for every such A, m, n?

$$\sum_{\mathcal{S}\subseteq\mathcal{D}} \det(m^{-1} H_{\mathcal{S}}) M(\mathcal{S}) x^{|\mathcal{S}|} = 0.$$

Why it matters

This would give an explicit algebraic relation for entries of a ubiquitous iterative matrix scaling limit, including a structured formula for every coefficient.

References

- E. Rowland and J. Wu, The entries of the Sinkhorn limit of an $m \times n$ matrix, arXiv v2 (2025-05-25), notation on pp. 3–4 and Conjecture 2. The matrix denoted $H_{\mathcal{S}}$ here is their $\mathrm{adj}_{\mathcal{S}}(m, n)$.
- E. Rowland, Combinatorial structure behind Sinkhorn limits, SIAM talk, 2025-07-07, slide 13: the coefficient formula remains conjectural after the degree bound was proved.
- M. C. Fang, Closed Form of a Generalized Sinkhorn Limit, 2025, abstract and the general algebraic degree bound. This settles the degree-bound consequence, rather than the displayed coefficient formula.

Status check — 2026-09-10

Rechecked Rowland–Wu v2, §2 and Conjecture 2, including the rectangular scaling and four coefficient signs, and Fang's degree-bound result. The 3×3 coefficient formula is proved, while the determinant formula in arbitrary dimensions remains conjectural. Searches for subsequent Rowland–Wu identity proofs found no full resolution. Fang's general degree bound alone does not establish the displayed coefficients.